

WFLOW-95 LAB: CMDB-Driven Host Tag Enrichment

Series: WFLOW — Workflows and Alert Notifications | **Reference:** 95 — CMDB-Driven Host Tag Enrichment LAB | **Created:** June 2026 | **Last Updated:** 09/28/2026

Overview

A frequent ask: read host → application mappings from a CMDB and apply them as host tags in bulk. This hands-on LAB builds that end to end as a Dynatrace **Workflow** — two *Run JavaScript* tasks that enrich hosts from CMDB lookup tables and apply the tags via the OneAgent Remote Configuration Management API, safely (dry-run first) and idempotently (skip tags that already exist).

It is the capstone for **WFLOW-08 (JavaScript & HTTP Actions)** — it combines an SDK DQL query (WFLOW-08 §2), `fetch()` to a Dynatrace API with auth (§3), retry logic (§4), `result()` / `withItems` data passing between tasks (§7), and the Credential Vault into one deployable workflow. Build it in the editor by following the [steps](#), or import the [YAML skeleton](#) at the end.

Is this the right tool? This LAB writes tags onto hosts through the classic remote configuration API. If the value can be derived from context the host already reports — host group, host name, an existing host tag — Latest Dynatrace has a simpler path: **Ingest enrichment configuration** (FAQ-02 § 3.3), a central rule with no host changes and no agent restart (OneAgent 1.343+). Use this LAB when the value exists **only in an external CMDB**, one value per host.

Table of Contents

1. [What This LAB Builds](#)

2. [The Two Tag Surfaces](#)
3. [Safety Model](#)
4. [Build It in the Workflows Editor](#)
5. [Prefer to Import? \(YAML Skeleton\)](#)
6. [Adapting It](#)
7. [Next Steps](#)
8. [Bonus — Seed Sample Lookup Tables \(testing only\)](#)

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Workflows (AutomationEngine) and Grail
Permissions	<code>automation:workflows:write</code> ; a Gen2 (classic) API token with <code>oneAgents.write</code> stored in the Credential Vault; and <code>environment-api:credentials:read</code> on the workflow's run-as identity (to read that token at runtime)
Data	Your CMDB exports uploaded as Grail lookup tables
Prior Knowledge	WFLOW-08 (JavaScript & HTTP actions); DQL <code>lookup</code> / <code>load</code>

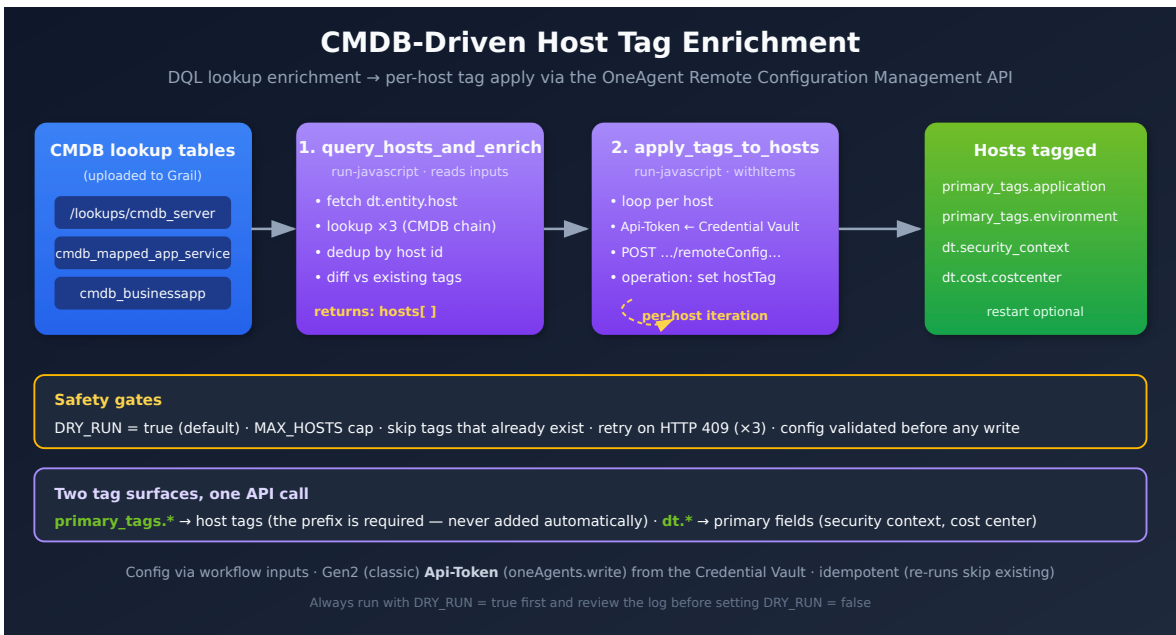
1. What This LAB Builds

The workflow is two `run-javascript` tasks chained by a loop:

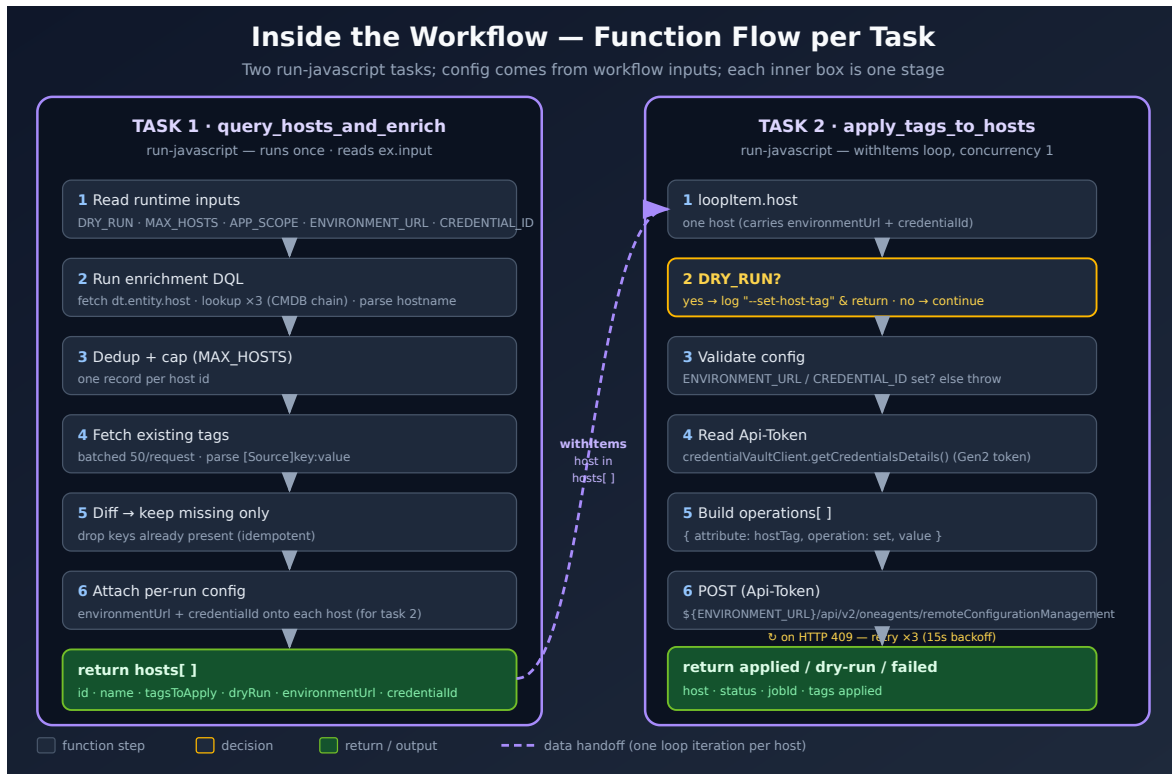
Task	Type	Role
<code>query_hosts_and_enrich</code>	<code>run-javascript</code>	Runs the DQL enrichment (host → three-table CMDB lookup chain), dedups by host id, fetches each host's current tags, and returns a <code>hosts[]</code> array containing only the tags that are missing .

Task	Type	Role
<code>apply_tags_to_hosts</code>	<code>run-javascript</code> with a loop	For each host in <code>hosts[]</code> , reads the API token from the Credential Vault and POSTs a <code>set hostTag</code> operation to the OneAgent Remote Configuration Management API — respecting dry-run and retrying on HTTP 409.

The write mechanism is `POST /api/v2/oneagents/remoteConfigurationManagement` with a `set hostTag` operation — the same `--set-host-tag` form you would run locally with `oneagentctl`, applied remotely across many hosts in one workflow.



The two tasks are chained: task 2 loops over `result('query_hosts_and_enrich').hosts` (one run per host) and only runs when task 1 found at least one host. The diagram below opens each task into its internal function flow.



2. The Two Tag Surfaces, One API Call

`TAG_MAPPING` maps an enrichment field to a host-tag key. Two different Dynatrace surfaces are written through the **same** `set hostTag` operation:

- `primary_tags.application` / `primary_tags.environment` become **primary tags**. The `primary_tags.` prefix is mandatory and is **never added for you** — a bare `application` key would create an ordinary tag, not a primary tag.
- `dt.security_context` and `dt.cost.costcenter` are **primary fields** (security context, cost-allocation), set through that same operation.

This split — and the primary-tags prefix rule — is covered in depth in **FAQ-02 (Tagging: Sources, Standards & Strategy)**. Tune `TAG_MAPPING` to the fields and tag keys your environment actually uses.

3. Safety Model

The workflow defaults to **safe**: a first run changes nothing. Read the dry-run log, confirm the proposed tags, then flip one flag.

Gate	What it does
<code>DRY_RUN = true</code> (default)	Task 2 only logs the <code>--set-host-tag</code> it <i>would</i> run — no write. Set to <code>false</code> to apply.
<code>MAX_HOSTS</code>	Caps hosts processed per run (default <code>50</code> ; <code>0</code> = unlimited).
Skip-existing	Task 1 reads current tags and drops any key already present — re-runs are idempotent.
409 retry	Remote Config Management allows one job per host at a time; task 2 retries up to <code>MAX_RETRIES</code> (3) with a 15s backoff on HTTP 409.
Credential Vault	Task 2 reads a Gen2 (classic) <code>oneAgents.write</code> API token from the vault at runtime via <code>credentialVaultClient</code> — never hardcoded. (The API reference now also lists a platform-token / OAuth scope, <code>fleet-management:oneagents:write</code> ; this LAB keeps the live-validated classic token until that path is re-tested.)
Config validation	Task 2 throws <i>before any live write</i> if <code>ENVIRONMENT_URL</code> / <code>CREDENTIAL_ID</code> still hold placeholders.
Loop concurrency <code>1</code>	Hosts are tagged one at a time (serialized loop) to avoid colliding jobs and API pressure.

`RESTART_ONEAGENT = false` means **"accepted", not "applied"**. The API reference states that *"By default OneAgents will be restarted when network zone, host group, host tags or host properties are reconfigured - the restart is required to apply the changes."* The LAB defaults to no restart so that a first live run cannot bounce agents unexpectedly — but with it off, a successful job (HTTP 2xx, a `jobId`) does not mean the tags are on the data yet; they apply when each host's OneAgent next restarts. Either schedule a restart window and set `RESTART_ONEAGENT = true`, or verify on data ingested after the hosts' next restart. Removals are slower still: *"Removing host properties and tags may require up to seven hours to take effect."*

4. Build It in the Workflows Editor (Step by Step)

Two *Run JavaScript* tasks wired with a loop. **All configuration lives in workflow inputs** (runtime variables) — task 1 reads them with `ex.input` and passes the values task 2

needs onto each host object, so the loop task needs no input wiring of its own.

UI labels vary between Workflows app versions. Where you define workflow inputs and the loop differs by version; the authoritative shape is the exported YAML (see the [import skeleton](#), which includes the `input:` block verbatim). The workflow is adapted from a working deployment, with its tenant URL and names genericized, rather than from a Dynatrace-published example — keep `DRY_RUN = true` for the first run in your tenant.

Step 0 — Prerequisites (once):

- **Upload your CMDB exports as Grail lookup tables** at the paths the query references (`/lookups/cmdb_server` , `/lookups/cmdb_mapped_app_service` , `/lookups/cmdb_businessapp`); adjust names/ fields to your schema. *(If they're missing, task 1's preflight stops the run cleanly with a "Missing CMDB lookup tables" message — no raw `UNKNOWN_TABULAR_FILE` error.)*
- **Create a Gen2 (classic) API token** with the `oneAgents.write` scope (`dt0c01...`), store it in the **Credential Vault**, and copy its credential ID (`CREDENTIALS_VAULT-...`).
- **Grant the workflow's run-as identity** `environment-api:credentials:read` so the workflow can read that token from the vault at runtime.

Why a classic token, not a platform token? The remote-config endpoint was live-validated as rejecting platform Bearer tokens at the scheme level (June 2026); the classic `Api-Token` path below is the proven approach. **Update (09/28/2026):** the API reference now documents platform-token support for this endpoint — *"Platform Token / OAuth: Required scope: fleet-management:oneagents:write"* (announced with SaaS 1.343, July 2026). This LAB keeps the validated classic-token flow until the platform-token path is re-tested live; verify in your tenant before switching.

Step 1 — Create the workflow and define its inputs. New workflow (leave the on-demand trigger). Define these **workflow inputs** — they become the parameters you can override in the Run dialog (easiest to set via the YAML editor — see the [import skeleton](#)):

Input	Default	Purpose
<code>DRY_RUN</code>	<code>true</code>	preview only; flip to <code>false</code> to apply

Input	Default	Purpose
MAX_HOSTS	50	cap per run; 0 = unlimited
APP_SCOPE	' '	limit to one business app; ' ' = all
EXCLUDE_PRODUCTION	false	skip environment == production
RESTART_ONEAGENT	false	restart OneAgent after tagging
ENVIRONMENT_URL	https://<your-environment>.live.dynatrace.com	tenant API base (no trailing slash)
CREDENTIAL_ID	CREDENTIALS_VAULT-...	vault entry holding the oneAgents.write token

Step 2 — Add the first JavaScript block. Click + under the trigger → **Run JavaScript**. Rename it `query_hosts_and_enrich`.

Step 3 — Fill it with this code. It preflights the lookup tables (so a missing upload fails clean), then enriches and diffs:

```

import { queryExecutionClient } from '@dynatrace-sdk/client-query';
import { execution } from '@dynatrace-sdk/automation-utils';

export default async function ({ execution_id }) {
  // ---- Runtime inputs (workflow `input` block; override per run) ----
  const ex = await execution(execution_id);
  const inp = ex.input || {};
  const MAX_HOSTS = Number(inp.MAX_HOSTS ?? 50);
  const DRY_RUN = (inp.DRY_RUN === false || inp.DRY_RUN === 'false') ? false :
true; // default true
  const RESTART_ONEAGENT = (inp.RESTART_ONEAGENT === true ||
inp.RESTART_ONEAGENT === 'true');
  const APP_SCOPE = (inp.APP_SCOPE ?? '').toString();
  const EXCLUDE_PRODUCTION = (inp.EXCLUDE_PRODUCTION === true ||
inp.EXCLUDE_PRODUCTION === 'true');
  const ENVIRONMENT_URL = (inp.ENVIRONMENT_URL ??
'').toString().replace(/\/+$/, '');
  const CREDENTIAL_ID = (inp.CREDENTIAL_ID ?? '').toString();

  const TAG_MAPPING = {
    'application': 'primary_tags.application',
    'environment': 'primary_tags.environment',
    'dt.security_context': 'dt.security_context',
    'dt.cost.costcenter': 'dt.cost.costcenter',
  };

  console.log('=== Host Tag Enrichment (runtime inputs) ===');
  console.log(`DRY_RUN=${DRY_RUN} | MAX_HOSTS=${MAX_HOSTS} |
APP_SCOPE='${APP_SCOPE}' | EXCLUDE_PRODUCTION=${EXCLUDE_PRODUCTION} |
RESTART_ONEAGENT=${RESTART_ONEAGENT}`);

  // ---- Preflight: confirm the CMDB lookup tables exist before querying
them, ----
  // ---- so a missing upload stops cleanly instead of throwing
UNKNOWN_TABULAR_FILE. ----
  const LOOKUP_TABLES = ['/lookups/cmdb_server',
'/lookups/cmdb_mapped_app_service', '/lookups/cmdb_businessapp'];
  const missingTables = await checkLookupTables(LOOKUP_TABLES);
  if (missingTables.length > 0) {
    console.log(`Missing CMDB lookup tables: ${missingTables.join(', ')}`);
    console.log('Upload them first (Step 0), then re-run. Stopping cleanly -
no hosts processed.');
```

```

    return { hosts: [], totalFound: 0, toProcess: 0, dryRun: DRY_RUN,
missingLookupTables: missingTables };
  }

  let enrichmentQuery = `fetch dt.entity.host

```

```

| fields id, entity.name, monitoringMode, paasVendorType,
isMonitoringCandidate, awsNameTag
| filter isMonitoringCandidate == false
| fieldsAdd monitoringMode = if(isNotNull(paasVendorType), "APP_ONLY",
else:if(isMonitoringCandidate == true, "CANDIDATE", else:monitoringMode)),
        provider = if(isNotNull(awsNameTag), "aws", else: "datacenter")
| filterOut monitoringMode == "APP_ONLY"
| fieldsRemove paasVendorType, awsNameTag, isMonitoringCandidate
| parse entity.name, ""LD:hostname ('.' LD:domain)? EOS""
| lookup [load "/lookups/cmdb_server"
        | fields cmdb_ci_key, name, dv_operational_status, environment,
busapp_cmdb_ci_key, apps_cmdb_ci_key], sourceField:hostname, lookupField:name,
prefix:"server."
| lookup [load "/lookups/cmdb_mapped_app_service"
        | fields environment, managed_by_email, cmdb_ci_key, busapp_name],
sourceField:server.apps_cmdb_ci_key, lookupField:cmdb_ci_key,
prefix:"mapped_app."
| lookup [load "/lookups/cmdb_businessapp"
        | fields short_name, owned_by, cmdb_ci_key],
sourceField:server.busapp_cmdb_ci_key, lookupField:cmdb_ci_key,
prefix:"bus_app."
| fields id, entity.name, application = lower(bus_app.short_name), environment
= lower(mapped_app.environment), dt.cost.costcenter =
lower(bus_app.short_name), dt.security_context = lower(bus_app.short_name)
| filter isNotNull(application) OR isNotNull(environment)`;

if (APP_SCOPE && APP_SCOPE.trim().length > 0) enrichmentQuery +=
`
| filter application == "${APP_SCOPE.trim().toLowerCase()}"`;
if (EXCLUDE_PRODUCTION) enrichmentQuery += `
| filterOut environment ==
"production"`;

const enriched = await executedqlQuery(enrichmentQuery);
console.log(`Found ${enriched.length} enriched host records`);

const hostMap = new Map();
enriched.forEach(record => {
  if (!hostMap.has(record.id)) {
    const hostData = { id: record.id, name: record['entity.name'],
application: record.application };
    Object.keys(TAG_MAPPING).forEach(f => { hostData[f] =
getNestedField(record, f); });
    hostMap.set(record.id, hostData);
  }
});
const uniqueHosts = Array.from(hostMap.values());
const hostsToProcess = MAX_HOSTS > 0 ? uniqueHosts.slice(0, MAX_HOSTS) :
uniqueHosts;
console.log(`${uniqueHosts.length} unique hosts; processing

```

```

    ${hostsToProcess.length}`);
    if (hostsToProcess.length === 0) return { hosts: [], totalFound: 0,
toProcess: 0, dryRun: DRY_RUN };

    const BATCH = 50;
    const existing = new Map();
    for (let i = 0; i < hostsToProcess.length; i += BATCH) {
        const batch = hostsToProcess.slice(i, i + BATCH);
        const filterParts = batch.map(h => `id == "${h.id}"`).join(' or ');
        const recs = await executeDqlQuery(`fetch dt.entity.host
| fieldsAdd tags
| filter ${filterParts}
| fields id, tags`);
        recs.forEach(record => {
            const allTags = (record.tags || []).filter(t => typeof t ===
'string');
            const keys = allTags.map(t => {
                let s = t; const b = t.lastIndexOf(''); if (b !== -1) s =
t.substring(b + 1);
                const c = s.indexOf(':'); return c === -1 ? null : s.substring(0, c);
            }).filter(k => k && Object.values(TAG_MAPPING).includes(k));
            existing.set(record.id, new Set(keys));
        });
    }

    const hostsWithTags = [];
    let skipped = 0;
    hostsToProcess.forEach(host => {
        const have = existing.get(host.id) || new Set();
        const tagsToApply = [];
        Object.entries(TAG_MAPPING).forEach(([f, tagKey]) => {
            const value = host[f];
            if (!value || String(value).trim().length === 0) return;
            if (have.has(tagKey)) { skipped++; return; }
            tagsToApply.push({ key: tagKey, value: sanitizeTagValue(value) });
        });
        if (tagsToApply.length > 0) {
            hostsWithTags.push({ id: host.id, name: host.name, application:
host.application, tagsToApply,
                dryRun: DRY_RUN, restartOneAgent: RESTART_ONEAGENT, environmentUrl:
ENVIRONMENT_URL, credentialId: CREDENTIAL_ID });
        }
    });

    console.log(DRY_RUN ? 'DRY RUN - no changes' : 'LIVE - will apply tags');
    hostsWithTags.forEach((h, i) => { console.log(`  ${i + 1}. ${h.name}
(${h.id})`; h.tagsToApply.forEach(t => console.log(`    +
${t.key}=${t.value}`)); });

```

```

    console.log(`Hosts to tag: ${hostsWithTags.length} | tags skipped (exist):
    ${skipped}`);

    return { hosts: hostsWithTags, totalFound: uniqueHosts.length, toProcess:
    hostsWithTags.length, totalSkipped: skipped, dryRun: DRY_RUN };
  }

  // Preflight helper: returns the subset of lookup-table paths that do NOT
  exist.
  async function checkLookupTables(paths) {
    const missing = [];
    for (const path of paths) {
      try {
        await executeDqlQuery(`load "${path}" | limit 1`);
      } catch (e) {
        if (e.message && e.message.includes('UNKNOWN_TABULAR_FILE'))
missing.push(path);
        else throw e; // a real/unexpected error - surface it
      }
    }
    return missing;
  }

  function getNestedField(record, fieldName) {
    if (record[fieldName] !== undefined) return record[fieldName];
    return fieldName.split('.').reduce((v, p) => (v == null ? undefined :
v[p]), record);
  }

  function sanitizeTagValue(value) { return String(value).replace(/\s/g,
'').trim().slice(0, 270); }

  async function executeDqlQuery(query) {
    let r = await queryExecutionClient.queryExecute({ body: { query,
requestTimeoutMilliseconds: 60000, maxResultRecords: 10000 } });
    while (r.state === 'RUNNING') { await new Promise(s => setTimeout(s,
1000)); r = await queryExecutionClient.queryPoll({ requestToken:
r.requestToken }); }
    if (r.state !== 'SUCCEEDED') throw new Error(`DQL failed: ${r.state}`);
    let recs = r.result?.records || [];
    while (r.result?.nextPageKey) { r = await queryExecutionClient.queryPoll({
requestToken: r.requestToken, nextPageKey: r.result.nextPageKey }); if
(r.result?.records) recs = recs.concat(r.result.records); }
    return recs;
  }
}

```

Step 4 — Customize task 1. Adjust `TAG_MAPPING`, the three `lookup [load`
`"/lookups/..."` tables and their `| fields ...` lists, and the final `| fields ... = lower(...)`
mapping to match your CMDB. **If you rename the lookup tables, update both the**

`LOOKUP_TABLES` preflight list and the `lookup [load ...]` calls. Run scope and behavior (`DRY_RUN`, `MAX_HOSTS`, etc.) come from the **inputs**, not code constants.

Step 5 — Add the second JavaScript block. Click **+** beneath task 1 → **Run JavaScript**.

Rename it `apply_tags_to_hosts` (placing it beneath task 1 makes task 1 its predecessor).

Step 6 — Fill it with this code:

```

import { actionExecution } from '@dynatrace-sdk/automation-utils';
import { credentialVaultClient } from '@dynatrace-sdk/client-classic-environment-v2';

const MAX_RETRIES = 3;
const RETRY_DELAY_MS = 15000; // 15s between retries on 409

export default async function ({ action_execution_id }) {
  const actionExe = await actionExecution(action_execution_id);
  const host = actionExe.loopItem.host;
  const ENVIRONMENT_URL = host.environmentUrl; // from task 1 (workflow input)
  const CREDENTIAL_ID = host.credentialId; // from task 1 (workflow input)

  console.log(`--- ${host.name} (${host.id}) - ${host.tagsToApply.length} tag(s) ---`);

  if (host.dryRun) {
    host.tagsToApply.forEach(t => console.log(` [DRY RUN] --set-host-tag=${t.key}=${t.value}`));
    return { host: host.name, hostId: host.id, status: 'dry-run', tagCount: host.tagsToApply.length };
  }

  if (!ENVIRONMENT_URL || ENVIRONMENT_URL.includes('&lt;')) throw new Error('ENVIRONMENT_URL input is not set.');
```

```

  if (!CREDENTIAL_ID || CREDENTIAL_ID.includes('X')) throw new Error('CREDENTIAL_ID input is not set.');
```

```

  // Gen2 (classic) API token with oneAgents.write, read from the Credential Vault.
  // Reading the vault needs environment-api:credentials:read on the run-as identity.
  const cred = await credentialVaultClient.getCredentialsDetails({ id: CREDENTIAL_ID });
  const API_TOKEN = cred.token;
  if (!API_TOKEN) throw new Error('Credential has no token value.');
```

```

  const operations = host.tagsToApply.map(tag => ({ attribute: 'hostTag', operation: 'set', value: `${tag.key}=${tag.value}` }));
  const url =
`${ENVIRONMENT_URL}/api/v2/oneagents/remoteConfigurationManagement?restart=${host.restartOneAgent}`;

  for (let attempt = 1; attempt <= MAX_RETRIES; attempt++) {
    try {
```

```

const response = await fetch(url, {
  method: 'POST',
  headers: { 'Authorization': `Api-Token ${API_TOKEN}`, 'Content-Type':
'application/json; charset=utf-8' },
  body: JSON.stringify({ entities: [host.id], operations })
});
if (response.status === 409 && attempt < MAX_RETRIES) { await
new Promise(s => setTimeout(s, RETRY_DELAY_MS)); continue; }
if (!response.ok) throw new Error(`HTTP ${response.status}: ${await
response.text()}`);
const data = await response.json();
console.log(` SUCCESS (attempt ${attempt}) jobId=${data.id || 'N/A'}`);
return { host: host.name, hostId: host.id, status: 'applied',
tagsApplied: host.tagsToApply.length, jobId: data.id || 'N/A', attempt };
} catch (error) {
  if (attempt < MAX_RETRIES && error.message &&
error.message.includes('409')) { await new Promise(s => setTimeout(s,
RETRY_DELAY_MS)); continue; }
  console.error(` FAILED (attempt ${attempt}): ${error.message}`);
  return { host: host.name, hostId: host.id, status: 'failed', error:
error.message, attempt };
}
}
return { host: host.name, hostId: host.id, status: 'failed', error: 'Max
retries exceeded' };
}

```

Step 7 — Customize task 2. Normally nothing: `ENVIRONMENT_URL` and `CREDENTIAL_ID` arrive from task 1 (sourced from the workflow inputs) on each host object, and the token is read from the vault at runtime. Just confirm Step 0's token and the `environment-api:credentials:read` grant.

Step 8 — Make task 2 loop over the hosts. Enable the loop (Options → Loop / "Run task for each item"):

- **Items:** `{{ result('query_hosts_and_enrich').hosts }}`
- **Loop item variable name:** `host` — must be `host` so `actionExe.loopItem.host` resolves.
- **Concurrency:** `1` (serialize the writes).

Run condition (Conditions → Custom): `{{ result('query_hosts_and_enrich').hosts | length > 0 }}`

Step 9 — Run with inputs. Save → Run. In the Run dialog set `ENVIRONMENT_URL` and `CREDENTIAL_ID` (leave `DRY_RUN = true`). Review the task-2 log's `--set-host-tag` lines,

then re-run with `DRY_RUN = false` to apply. Add a schedule trigger (WFLOW-02) to keep newly onboarded hosts tagged; skip-existing keeps repeat runs cheap and idempotent.

5. Prefer to Import? (YAML Skeleton)

Build the envelope below — it includes the `input:` block and the loop/condition wiring — paste the **Step 3** and **Step 6** scripts into the `script:` blocks, and import it from the Workflows app.

```

metadata:
  version: '1'
  dependencies:
    apps:
      - id: dynatrace.automations
        version: ^1.3074.2
workflow:
  title: Host Tag Enrichment from Lookup Table
  schemaVersion: 4
  trigger: {}
  type: STANDARD
  input:
    DRY_RUN: true
    APP_SCOPE: ''
    MAX_HOSTS: 50
    ENVIRONMENT_URL: 'https://&lt;your-environment&gt;.live.dynatrace.com'
    CREDENTIAL_ID: CREDENTIALS_VAULT-XXXXXXXXXXXXXXXXXX
    RESTART_ONEAGENT: false
    EXCLUDE_PRODUCTION: false
  tasks:
    query_hosts_and_enrich:
      name: query_hosts_and_enrich
      action: dynatrace.automations:run-javascript
      predecessors: []
      input:
        script: |
          # paste the Step 3 script (query_hosts_and_enrich) here
    apply_tags_to_hosts:
      name: apply_tags_to_hosts
      action: dynatrace.automations:run-javascript
      predecessors:
        - query_hosts_and_enrich
      conditions:
        custom: '{{ result(''query_hosts_and_enrich'').hosts | length &gt; 0
}}'
      states:
        query_hosts_and_enrich: OK
      withItems: host in {{ result('query_hosts_and_enrich').hosts }}
      concurrency: 1
      input:
        script: |
          # paste the Step 6 script (apply_tags_to_hosts) here

```

6. Adapting It

- **Runtime inputs** — every knob (`DRY_RUN` , `MAX_HOSTS` , `APP_SCOPE` , `EXCLUDE_PRODUCTION` , `RESTART_ONEAGENT` , `ENVIRONMENT_URL` , `CREDENTIAL_ID`) is a workflow input, overridable per run in the Run dialog. No code edits to change scope or target.
- **Scope filters** — `APP_SCOPE` limits the run to one business app; `EXCLUDE_PRODUCTION` skips hosts whose enriched `environment` is `production` . Both are appended to the DQL at runtime.
- `sanitizeTagValue` strips whitespace and caps values at 270 characters (Dynatrace tag-value constraint).
- **Schedule it** with a cron trigger (WFLOW-02) so newly onboarded hosts get tagged automatically; skip-existing keeps repeat runs cheap.
- **Token type** — this uses a **Gen2 (classic)** `Api-Token` , the live-validated path. The API reference now lists `fleet-management:oneagents:write` as the platform-token / OAuth scope for this endpoint (read 09/28/2026) — once verified in your tenant, the call can move to platform-token auth, potentially with no stored token at all if the workflow's run-as identity carries the scope.

7. Next Steps

- **WFLOW-02: Triggers** — add a schedule trigger to run this enrichment on a cadence.
- **WFLOW-07: Auto-Remediation** — the `DRY_RUN` / max-attempts / serialized-write guardrails here are the same safety patterns used for remediation tasks.
- **WFLOW-08: JavaScript & HTTP Actions** — the building blocks this LAB composes (SDK DQL, `fetch()` , `retry`, `result()` / `withItems` , Credential Vault).
- **FAQ-02: Tagging — Sources, Standards & Strategy** — the tagging-source model and the `primary_tags.` prefix rule that this workflow writes against.

References

- [OneAgent remote configuration management API — POST a configuration job \(DT docs\)](#)
- [Remote configuration management of OneAgents and ActiveGates \(DT docs\)](#)

- [Lookup data in Grail \(DT docs\)](#)
- [Build workflows \(DT docs\)](#)
- [Run JavaScript action \(DT docs\)](#)
- [OneAgent tag setup — primary tags / fields \(DT docs\)](#)

Sources: [OneAgent remote configuration management API — POST a configuration job \(DT docs\)](#) — documents the `hostTag set` operation, the `dt.cost.costcenter=<value>` form, and the `oneAgents.write` scope, [Build workflows \(DT docs\)](#), [Lookup data in Grail \(DT docs\)](#), [OneAgent tag setup \(DT docs\)](#).

Appendix — Bonus: Seed Sample Lookup Tables (Testing Only)

Not part of the main workflow. In production the three CMDB lookup tables are uploaded from your **real CMDB** (Step 0). This appendix is only for trying the LAB on a tenant that has **no CMDB tables** — it creates three tiny sample tables so the enrichment resolves for one host. **Do not run it where the real tables already exist** (it would overwrite them).

It uses the Grail **Resource Store API** (`/platform/storage/resource-store/v1/files/tabular/lookup:upload`) — the reverse direction of this LAB (writing to a lookup table instead of reading *from* one). The same API is how you'd build a real "sync CMDB → lookup table" job if your CMDB is reachable from a workflow.

Prerequisite: the workflow's run-as identity needs `storage:files:write` (reading the tables back in DQL needs nothing extra).

How to use: create a **separate** one-task workflow (don't add this to the tagging workflow), paste the code, **replace** `myhost-01` with the short hostname of a real host in your tenant (the part before the first dot), and run it once. Then verify with `load "/lookups/cmdb_server" | limit 10` and run the main workflow with `DRY_RUN = true`. The sample chain resolves to `application = sampleapp`, `environment = dev`.

```

const BASE = '/platform/storage/resource-store/v1/files/tabular';

const TABLES = [
  {
    filePath: '/lookups/cmdb_server', lookupField: 'name', displayName:
'cmdb_server',
    parsePattern: 'JSON{STRING:name, STRING:cmdb_ci_key,
STRING:dv_operational_status, STRING:environment, STRING:busapp_cmdb_ci_key,
STRING:apps_cmdb_ci_key}(flat=true)',
    rows: [{ name: 'myhost-01', cmdb_ci_key: 'SRV1', dv_operational_status:
'operational',
      environment: 'dev', busapp_cmdb_ci_key: 'BAPP1',
apps_cmdb_ci_key: 'APP1' }],
  },
  {
    filePath: '/lookups/cmdb_mapped_app_service', lookupField: 'cmdb_ci_key',
displayName: 'cmdb_mapped_app_service',
    parsePattern: 'JSON{STRING:cmdb_ci_key, STRING:environment,
STRING:managed_by_email, STRING:busapp_name}(flat=true)',
    rows: [{ cmdb_ci_key: 'APP1', environment: 'dev',
      managed_by_email: 'owner@example.com', busapp_name: 'Sample App'
}],
  },
  {
    filePath: '/lookups/cmdb_businessapp', lookupField: 'cmdb_ci_key',
displayName: 'cmdb_businessapp',
    parsePattern: 'JSON{STRING:cmdb_ci_key, STRING:short_name,
STRING:owned_by}(flat=true)',
    rows: [{ cmdb_ci_key: 'BAPP1', short_name: 'sampleapp', owned_by:
'platform-team' }],
  },
];

export default async function () {
  const out = {};
  for (const t of TABLES) {
    const jsonl = t.rows.map(r => JSON.stringify(r)).join('\n');

    // delete first (ignore "does not exist") so re-runs are clean
    try {
      const del = await fetch(`${BASE}/lookup:delete`, { method: 'POST', body:
JSON.stringify({ filePath: t.filePath }) });
      console.log(`[delete] ${t.filePath} -> HTTP ${del.status}`);
    } catch (e) { console.log(`[delete] ${t.filePath} skipped: ${e.message}`);
    }

    // upload (multipart: request as a JSON string + content blob; NO headers)

```

```

const form = new FormData();
form.append('request', JSON.stringify({
  filePath: t.filePath,
  displayName: t.displayName,
  description: 'Sample CMDB data for testing host-tag enrichment',
  parsePattern: t.parsePattern,
  skippedRecords: 0,
  overwrite: true,
  lookupField: t.lookupField,
  timezone: 'UTC',
  locale: 'en_US',
}));
form.append('content', new Blob([jsonl], { type: 'text/plain' }),
'data.jsonl');

const resp = await fetch(`${BASE}/lookup:upload`, { method: 'POST', body:
form });
const body = await resp.text();
console.log(`[upload] ${t.filePath} -> HTTP ${resp.status} ::
${body.slice(0, 300)}`);
out[t.filePath] = { status: resp.status, ok: resp.ok };
if (!resp.ok) console.error(`Upload failed for ${t.filePath}: ${body}`);
}
console.log('Verify each: load "/lookups/cmdb_server" | limit 10');
return out;
}

```

Upload mechanics (reusable for any lookup-table upload):

- `parsePattern: 'JSON{STRING:field, INT:field, ...}(flat=true)'` produces **flat top-level columns** from JSONL — bare `JSON` would nest them under one field.
- The `request` part is a plain **JSON string**; `content` is a `Blob`; **set no headers** — `fetch` builds the multipart boundary.
- The **relative URL** is authenticated by the platform as the workflow's run-as identity — **no token needed**. (This is a platform-native endpoint, unlike the classic remote-config API the main LAB uses, which requires a Gen2 Api-Token.)
- **Re-runnable**: it deletes then re-uploads. To remove a table, POST `{"filePath": "/lookups/<name>"} to .../lookup:delete.`

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.